

1 $f(x) = x + 2$.

Find the values of:

a $f(f(0))$

4

b $f(f(3))$

7

c $f(f(-1))$

3

d $f(f(-8))$

-4

e Find an expression for $f(f(x))$

$x + 4$

f Check your answers for **a – d**, how do they relate to your expression for $f(f(x))$?

The composite function is to add 4 to the value of x , which reflect the answers to part a – d.

g Find a value of x where $f(f(x)) = 0$.

$x = -4$

2 $f(x) = 2x$

Find the values of:

a $f(f(0))$

0

b $f(f(7))$

28

c $f(f(-3))$

-12

d $f(f(-11))$

-44

e Find an expression for $f(f(x))$

$4x$

f Check your answers for **a – d**, how do they relate to your expression for $f(f(x))$?

The composite function is to multiply the value of x by 4, which reflect the answers to parts a – d.

g Find a value of x where $f(f(x)) = 32$.

$x = 8$

3 $g(x) = 2 - x$. Show that $g(g(x)) = x$.

$$\begin{aligned} g(x) &= 2 - x \\ g(2 - x) &= 2 - (2 - x) \\ \therefore g(2 - x) &= x \end{aligned}$$

4 $h(x) = 3x - 5$. Find an expression for $h(h(x))$.

$$9x - 20$$

5 Let $f(x) = 3x + 4$ and $g(x) = 3 - 2x$.

a Find $f(g(2))$

b Find $g(f(2))$

c Show that $f(g(x)) = -6x + 13$

d Show that $g(f(x)) = -6x - 5$

$$\begin{aligned} f(x) &= 3x + 4 \\ f(g(x)) &= f(3 - 2x) \\ &= 3(3 - 2x) + 4 \\ &= -6x + 13 \end{aligned}$$

$$\begin{aligned} g(x) &= 3 - 2x \\ g(f(x)) &= g(3x + 4) \\ &= 3 - 2(3x + 4) \\ &= -6x - 5 \end{aligned}$$

6 $f(x) = x + 1$ and $g(x) = 2x - 3$

Find the values of:

a $f(g(7))$

b $g(f(7))$

c $f(f(7))$

d $g(g(7))$

$$12$$

$$13$$

$$9$$

$$19$$

Find expressions for

e $f(g(x))$

f $g(f(x))$

g $f(f(x))$

h $g(g(x))$

$$2x - 2$$

$$2x - 1$$

$$x + 2$$

$$4x - 9$$

7 Given $f(x) = 2x + 7$ and $g(x) = 5x - 3$, find:

a $f(g(x))$

b $g(f(x))$

$$10x + 1$$

$$10x + 32$$

c $f(f(x))$

d $g(g(x))$

$$4x + 21$$

$$25x - 18$$

8 $f(x) = \sqrt{x}$ and $g(x) = x + 4$

a Find $f(g(x))$

b Find $g(f(x))$

$$\sqrt{x+4}$$

$$\sqrt{x} + 4$$

c Find $g(g(x))$

d What composite function would produce $\sqrt{x+8}$?

$$x + 8$$

$$f(g(g(x)))$$

9 Given $p(x) = x^2 - 4$ and $q(x) = 2x + 3$, simplify:

a $p(q(x))$

b $q(p(x))$

$$4x^2 + 12x + 5$$

$$2x^2 - 5$$

c $p(p(x))$

d $q(q(x))$

$$x^4 - 8x^2 + 12$$

$$4x + 9$$

10 Given $h(x) = \frac{2}{x+3}$ and $m(x) = 4x - 1$, find:

a $h(m(x))$

b $m(h(x))$

c $h(h(x))$

$$\frac{1}{2x+1}$$

$$\frac{5-x}{x+3}$$

$$\frac{2x+6}{3x+11}$$

11 Find $f(g(x))$ and $g(f(x))$ for each of these function pairs.

a $f(x) = x - 2, g(x) = 2x + 5$ $f(g(x))$ $g(f(x))$	b $f(x) = 2x, g(x) = 3x$ $f(g(x))$ $g(f(x))$
c $f(x) = x^2, g(x) = x^3$ $f(g(x))$ $g(f(x))$	d $f(x) = \frac{1}{x}, g(x) = \sqrt{x}$ $f(g(x))$ $g(f(x))$

12 $f(x) = \frac{x}{x+1}$, find $f(f(x))$.

13 HSC Sample Question Band 4

Given $f(2) = 5, f(3) = 4, g(2) = 5, g(3) = 2, g(4) = 1$.

What is the value of $f(g(3))$?

$f(g(3)) = f(2) = 5$

14 HSC Sample Question Band 4

Which of these functions satisfies $f(f(x)) = x$?

- A. $f(x) = 2 - x$
- B. $f(x) = x^2$
- C. $f(x) = 2\sqrt{x}$
- D. $f(x) = x - 2$

A

15 HSC Sample Question Band 4

Which of these functions satisfies $f(f(x)) = x$?

- A. $f(x) = x + 1$
- B. $f(x) = x - 1$
- C. $f(x) = \frac{x - 1}{x + 1}$
- D. $f(x) = \frac{x + 1}{x - 1}$

D

16 Consider the graphs of $f(x)$ and $g(x)$ below.

a Evaluate $g(3)$.

b Hence evaluate $f(g(3))$.

c Evaluate $g(f(0))$.

d $f(f(1))$.

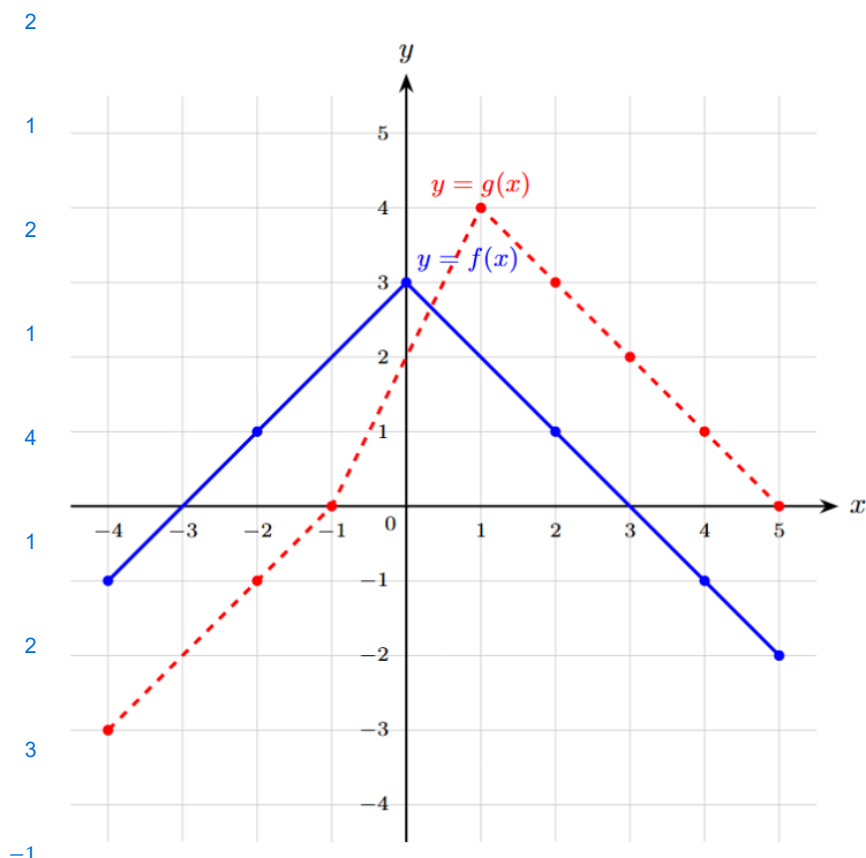
e $g(g(4))$.

f $f(g(3))$

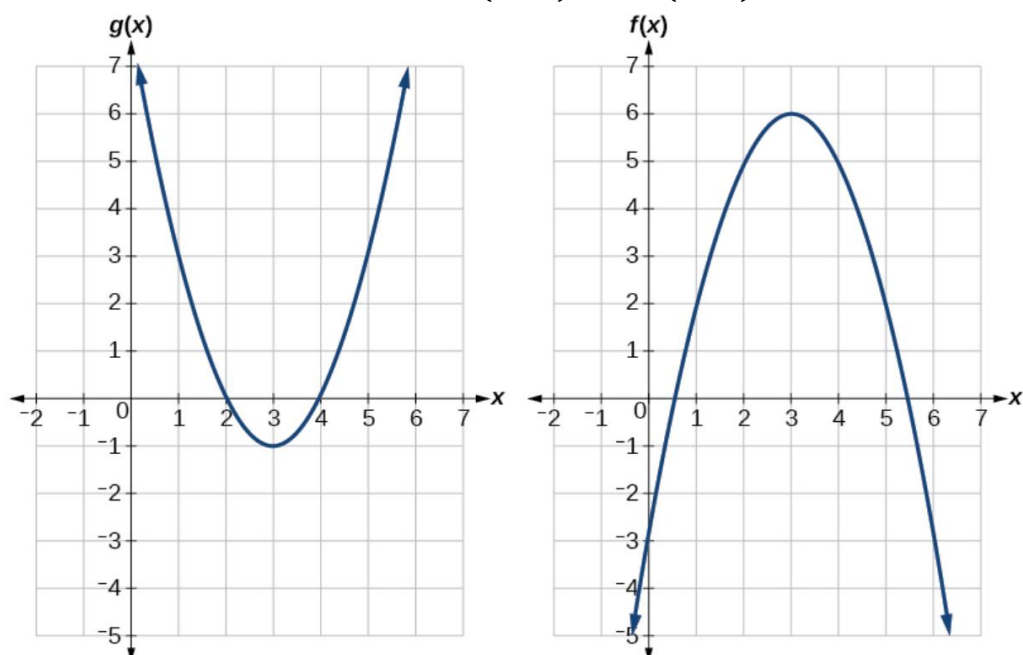
g $g(f(3))$

h $f(g(5))$

i $g(f(5))$



17 Using the graphs below, evaluate $f(g(1))$ and $g(f(2))$



$$f(g(1)): g(1) = 3, f(3) = 6$$

$$g(f(2)): f(2) = 5, g(5) = 3$$

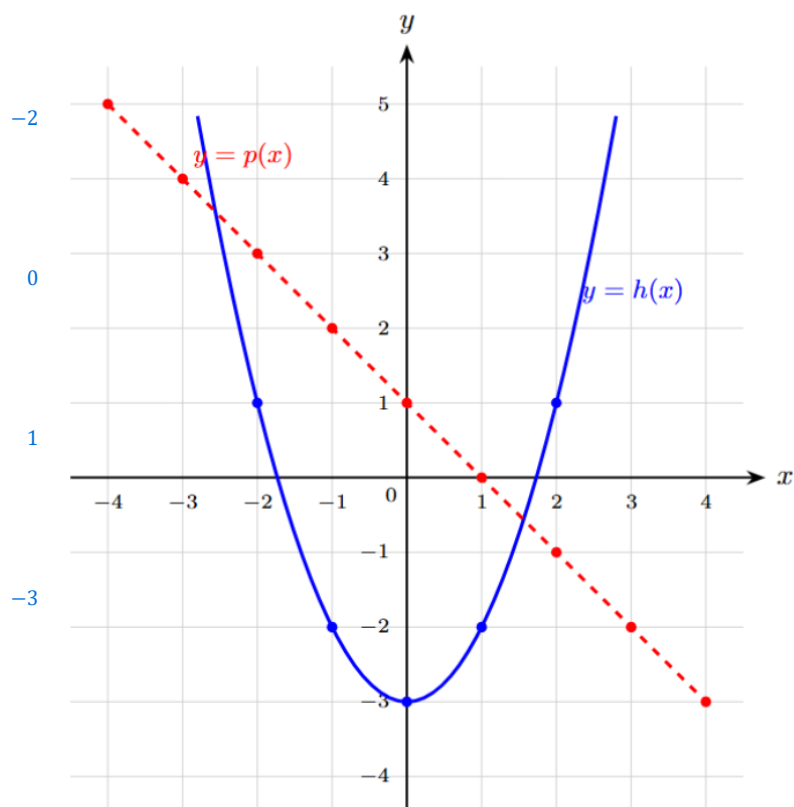
18 Consider the graphs of $p(x)$ and $h(x)$ below. Evaluate:

a $h(p(2))$

b $p(h(2))$

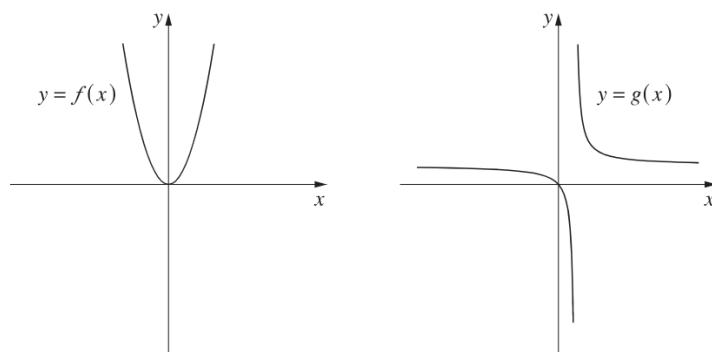
c $h(h(1))$

d $p(p(-3))$

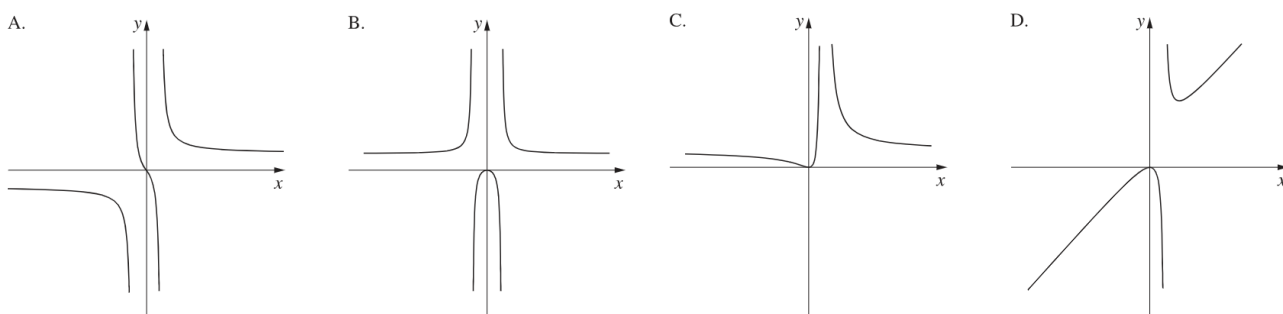


19 2022 HSC Advanced Band 5

The graphs of $y = f(x)$ and $y = g(x)$ are shown.



Which graph best represents $y = g(f(x))$?

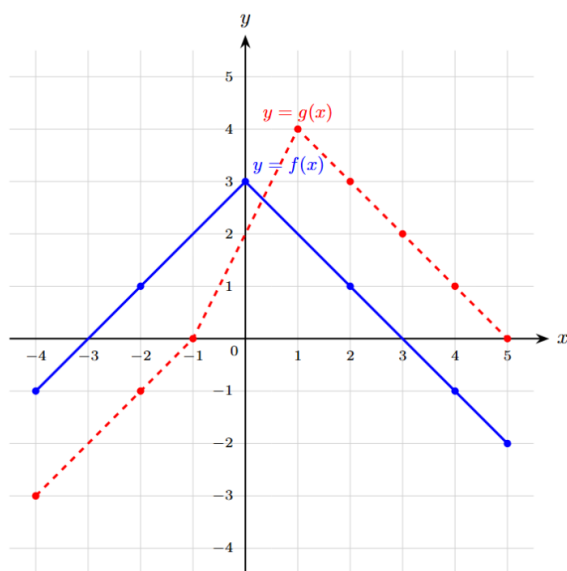


Since $f(x)$ is even, $g(f(x))$ will also be even.

B

20 Consider once again the graph of $f(x)$ and $g(x)$ below.

- a** Find all values of x in the domain $[-4,5]$ that satisfy $f(g(x)) = 3$.



Consider $f(g(x)) = 3$

Look for points where graph f reaches height $y = 3$.

From the grid, $f(g(x)) = 3$ only when $g(x) = 0$.

Solve $g(x) = 0$:

$x = -1, 5$

- b** [2-cycle fixed points]

- Evaluate $f f f f(x)$ for $x = 0$.
- Evaluate $f f f f(x)$ for $x = 3$.
- Explain geometrically why the pair $\{0,3\}$ forms an alternating cycle under composition by f , and state other alternating cycles for f .

Repeated compositions of $x = 0$ and $x = 3$ keep alternating between 0 and 3.

An alternating cycle occurs when $f(a) = b$ and $f(b) = a$

Geometrically, this means (a, b) and (b, a) must lie on the graph of $y = f(x)$

These two points are mirror reflections across the line $y = x$

Therefore, all points with symmetry across $y = x$ on the graph of $f(x)$ will have this property, such as $\{1,2\}$ and $\{0.5,2.5\}$

By considering the graph, the solution set is all (x, y) pairs satisfying $y = 3 - x$ in the domain $x \in [0,3]$

1 For each function, find its inverse $f^{-1}(x)$ then verify $f(f^{-1}(x)) = x$ and $f^{-1}(f(x)) = x$.

a $f(x) = x^3 + 1$

Verify $f(f^{-1}(x)) = x$ $f^{-1}(x) = \sqrt[3]{x-1}$

Verify $f^{-1}(f(x)) = x$

b $f(x) = \frac{1-x}{3+x}$

Verify $f(f^{-1}(x)) = x$ $f^{-1}(x) = \frac{1-3x}{1+x}$

Verify $f^{-1}(f(x)) = x$

c $f(x) = \sqrt[3]{x-7}$

Verify $f(f^{-1}(x)) = x$ $f^{-1}(x) = x^3 + 7$

Verify $f^{-1}(f(x)) = x$

2 $f(x) = 3x + 4$ and $g(x) = ax^2$

a Work out an expression for $f^{-1}(x)$

b Given that $fg(3) = 85$, find the value of a .

$$f^{-1}(x) = \frac{x-4}{3}$$

$$g(3) = 9a$$

$$f(9a) = 27a + 4 = 85$$

$$a = 3$$

- 3** $f(x) = ax + 8$ and $g(x) = 2x - 3$
- a** Work out an expression for $f^{-1}(x)$
- b** Given $fg(2) = 13$, find the value of a .

$$f^{-1}(x) = \frac{x-8}{a}$$

$$g(2) = 1$$

$$f(1) = a + 8 = 13$$

$$a = 5$$

- 4** $f(x) = x^2 - 1$ and $g(x) = 4x + 5$
- a** Work out an expression for $f^{-1}(x)$
- b** Work out an expression for $g^{-1}(x)$
- c** Given that $fg(x) = 3gf(x)$, show that $4x^2 + 40x + 21 = 0$

$$f^{-1}(x) = \sqrt{x+1}, \quad g^{-1}(x) = \frac{x-5}{4}$$

$$fg(x) = (4x+5)^2 - 1 = 16x^2 + 40x + 24$$

$$gf(x) = 4(x^2 - 1) + 5 = 4x^2 + 1$$

$$16x^2 + 40x + 24 = 3(4x^2 + 1)$$

$$4x^2 + 40x + 21 = 0$$

- 5** $f(x) = 2x + 4$ and $g(x) = 3x - 8$
- a** Find $f^{-1}(x)$ and $g^{-1}(x)$
- b** Find an expression for $fg(x)$
- c** Find an expression for the inverse of $fg(x)$
- d** Is $(fg)^{-1}(x)$ the same as $f^{-1}g^{-1}(x)$?
- e** Is $(fg)^{-1}(x)$ the same as $g^{-1}f^{-1}(x)$?

$$f^{-1}(x) = \frac{x-4}{2}, \quad g^{-1}(x) = \frac{x+8}{3}$$

$$fg(x) = 2(3x-8) + 4 = 6x - 12$$

$$(fg)^{-1}(x) = \frac{x+12}{6}$$

$$f^{-1}g^{-1}(x) = \frac{x-4}{6} \quad \text{not the same as } (fg)^{-1}(x)$$

$$g^{-1}f^{-1}(x) = \frac{x+12}{6} \quad \text{yes, the same as } (fg)^{-1}(x)$$

6

- a Let $f(x)$ be the function “increase x by 50%”. Write the equation for $f(x)$.

$$f(x) = 1.5x$$

- b A student in Year 8 says:



If you increase by 50%, then you can get back to the original by decreasing by 50%.

Let $g(x)$ be “decrease x by 50%.”

- i Write the equation for $g(x)$.

$$g(x) = 0.5x$$

- ii Show that she is incorrect by finding $g(f(x))$.

$$g(1.5x) = 0.5(1.5x) = 0.75x$$

Since $g(f(x)) \neq x$, $g(x)$ is not the inverse of $f(x)$

Therefore, a decrease of 50% will not undo an increase of 50%

- c What is the inverse function $f^{-1}(x)$?

$$f^{-1}(x) = \frac{x}{1.5} = \frac{2x}{3}$$

- d Therefore, to undo a 50% increase, what percentage do you need to decrease by?

$$f^{-1}(x) = \frac{2}{3}x, \text{ we need to decrease } x \text{ by a third,}$$

so 33.3% decrease

7

- a Show that $\frac{x^2+x}{x^2+4x+3}$ can be written as $\frac{x}{x+a}$ where a is a positive integer.

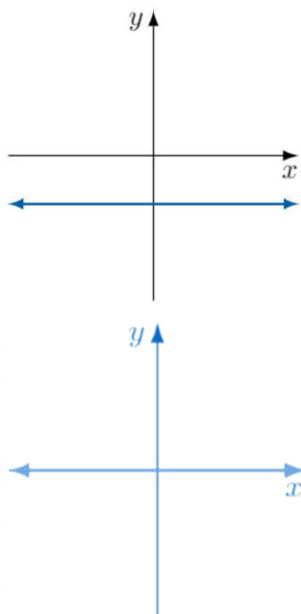
$$\frac{x(x+1)}{(x+1)(x+3)} = \frac{x}{x+3}, \quad a = 3$$

- b Hence, given $f(x) = \frac{x^2+x}{x^2+4x+3}$, find an expression for $f^{-1}(x)$

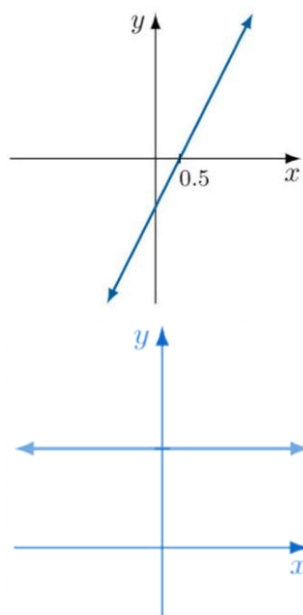
$$f^{-1}(x) = \frac{3x}{1-x}$$

1 Sketch the derivative function for each curve.

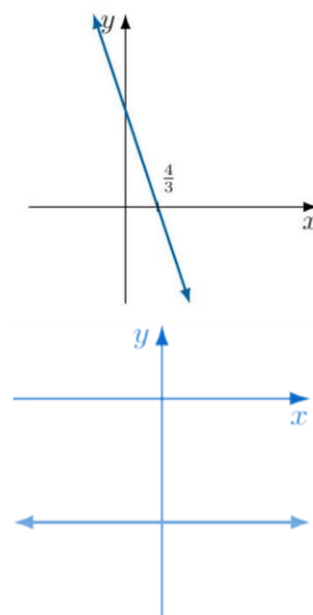
a



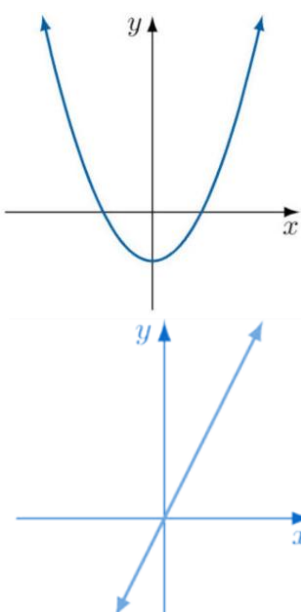
b



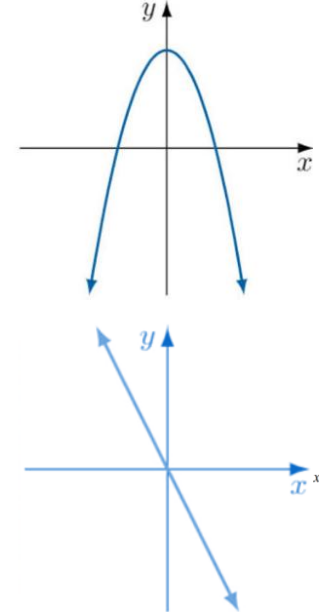
c



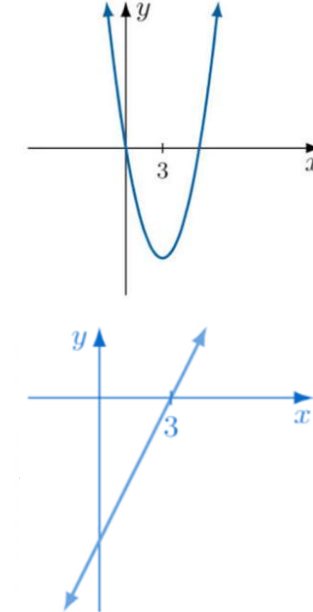
d



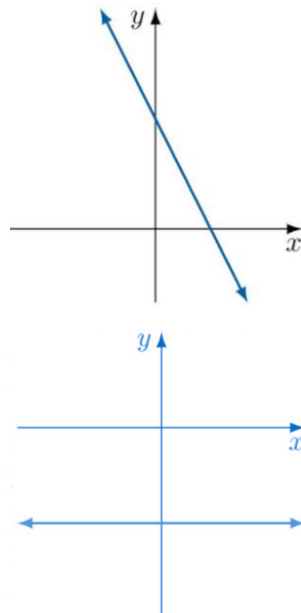
e



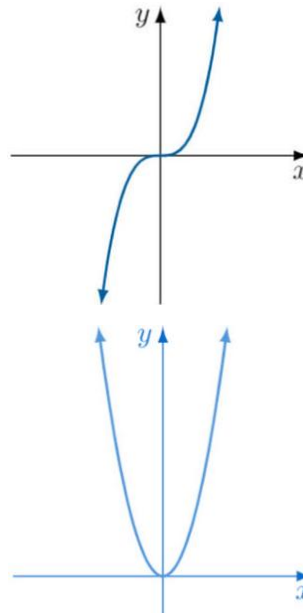
f



g



h



i

